PL/SQL

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Experiment Title: Implementation of PL/SQL Commands

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PL/SQL LAB PROGRAMS

EXERCISE 8 – PL/SQL BASICS

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1. Write a simple PL/SQL Block to print Hello World.

```
Connected.
SQL> SET SERVEROUTPUT ON;
SQL> -- Q1: Print Hello World
SQL>
SQL> BEGIN
  2      DBMS_OUTPUT.PUT_LINE('Hello World');
  3  END;
  4  /
Hello World

PL/SQL procedure successfully completed.
```

---

2. Accept two numbers and perform four arithmetic operations

```
SQL> -- Q2: Accept two numbers and perform arithmetic operations
SQL>
SQL> DECLARE
  2      a NUMBER;
  3      b NUMBER;
  4  BEGIN
  5      a := &a;
  6      b := &b;
  7
  8      DBMS_OUTPUT.PUT_LINE('Addition = ' || (a+b));
  9      DBMS_OUTPUT.PUT_LINE('Subtraction = ' || (a-b));
 10      DBMS_OUTPUT.PUT_LINE('Multiplication = ' || (a*b));
 11      DBMS_OUTPUT.PUT_LINE('Division = ' || (a/b));
 12  END;
 13  /
Enter value for a: 55
old   5:     a := &a;
new   5:     a := 55;
Enter value for b: 5
old   6:     b := &b;
new   6:     b := 5;
Addition = 60
Subtraction = 50
Multiplication = 275
Division = 11
```

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3. Fetch salary of employee named Ravi

```
SQL> -- Q3: Fetch salary of employee Ravi
SQL>
SQL> DECLARE
  2      v_salary EMPLOYEE_24BYB1117.SALARY%TYPE;
  3  BEGIN
  4      SELECT SALARY
  5      INTO v_salary
  6      FROM EMPLOYEE_24BYB1117
  7      WHERE EMP_NAME = 'Anand';
  8
  9      DBMS_OUTPUT.PUT_LINE('Salary of Anand is ' || v_salary);
 10  END;
 11  /
Salary of Anand is 45000

PL/SQL procedure successfully completed.
```

---

4. Accept department number and print employee count

```
SQL> -- Q4: Count employees in a department
SQL>
SQL> DECLARE
  2      v_dept VARCHAR2(20);
  3      v_count NUMBER;
  4  BEGIN
  5      v_dept := '&dept';
  6
  7      SELECT COUNT(*)
  8      INTO v_count
  9      FROM EMPLOYEE_24BYB1117
 10      WHERE DEPT = v_dept;
 11
 12      DBMS_OUTPUT.PUT_LINE('Employee count = ' || v_count);
 13  END;
 14  /
Enter value for dept: IT
old   5:     v_dept := '&dept';
new   5:     v_dept := 'IT';
Employee count = 2

PL/SQL procedure successfully completed.
```

---

5. Check voting eligibility

```
SQL> -- Q5: Check if person is eligible to vote
SQL>
SQL> DECLARE
  2      age NUMBER;
  3  BEGIN
  4      age := &age;
  5
  6      IF age ≥ 18 THEN
  7          DBMS_OUTPUT.PUT_LINE('Eligible to Vote');
  8      ELSE
  9          DBMS_OUTPUT.PUT_LINE('Not Eligible to Vote');
 10      END IF;
 11  END;
 12  /
Enter value for age: 26
old   4:     age := &age;
new   4:     age := 26;
Eligible to Vote

PL/SQL procedure successfully completed.
```

---

6. Find greatest of three numbers

```
SQL> -- Q6: Find greatest of three numbers
SQL>
SQL> DECLARE
  2      a NUMBER := &a;
  3      b NUMBER := &b;
  4      c NUMBER := &c;
  5  BEGIN
  6      IF a>b AND a>c THEN
  7          DBMS_OUTPUT.PUT_LINE('Greatest number is '||a);
  8      ELSIF b>a AND b>c THEN
  9          DBMS_OUTPUT.PUT_LINE('Greatest number is '||b);
 10      ELSE
 11          DBMS_OUTPUT.PUT_LINE('Greatest number is '||c);
 12      END IF;
 13  END;
 14  /
Enter value for a: 3
old   2:      a NUMBER := &a;
new   2:      a NUMBER := 3;
Enter value for b: 5
old   3:      b NUMBER := &b;
new   3:      b NUMBER := 5;
Enter value for c: 6
old   4:      c NUMBER := &c;
new   4:      c NUMBER := 6;
Greatest number is 6
```

---

7. Print day of week

```
SQL> -- Q7: Input number (1-7) and print day
SQL>
SQL> DECLARE
  2      d NUMBER := &day;
  3  BEGIN
  4      CASE d
  5          WHEN 1 THEN DBMS_OUTPUT.PUT_LINE('Sunday');
  6          WHEN 2 THEN DBMS_OUTPUT.PUT_LINE('Monday');
  7          WHEN 3 THEN DBMS_OUTPUT.PUT_LINE('Tuesday');
  8          WHEN 4 THEN DBMS_OUTPUT.PUT_LINE('Wednesday');
  9          WHEN 5 THEN DBMS_OUTPUT.PUT_LINE('Thursday');
 10          WHEN 6 THEN DBMS_OUTPUT.PUT_LINE('Friday');
 11          WHEN 7 THEN DBMS_OUTPUT.PUT_LINE('Saturday');
 12          ELSE DBMS_OUTPUT.PUT_LINE('Invalid Day');
 13      END CASE;
 14  END;
 15  /
Enter value for day: 5
old   2:      d NUMBER := &day;
new   2:      d NUMBER := 5;
Thursday

PL/SQL procedure successfully completed.
```

---

8. Sum of first N natural numbers using FOR loop

```
SQL> -- Q8: Sum of first n natural numbers using FOR loop
SQL>
SQL> DECLARE
  2      n NUMBER := &n;
  3      s NUMBER := 0;
  4  BEGIN
  5      FOR i IN 1..n LOOP
  6          s := s + i;
  7      END LOOP;
  8
  9      DBMS_OUTPUT.PUT_LINE('Sum = ' || s);
 10  END;
 11  /
Enter value for n: 4
old   2:     n NUMBER := &n;
new   2:     n NUMBER := 4;
Sum = 10

PL/SQL procedure successfully completed.
```

---

9. Sum of first N natural numbers using WHILE loop

```
SQL> -- Q9: Sum using WHILE loop
SQL>
SQL> DECLARE
  2      n NUMBER := &n;
  3      i NUMBER := 1;
  4      s NUMBER := 0;
  5  BEGIN
  6      WHILE i≤n LOOP
  7          s := s + i;
  8          i := i + 1;
  9      END LOOP;
 10
 11      DBMS_OUTPUT.PUT_LINE('Sum = ' || s);
 12  END;
 13  /
Enter value for n: 66
old   2:     n NUMBER := &n;
new   2:     n NUMBER := 66;
Sum = 2211

PL/SQL procedure successfully completed.
```

EXERCISE 9 – PROCEDURES AND FUNCTIONS

---

10. PL/SQL function to print Hello World

```
SQL> -- Q10: Function to print Hello World
SQL>
SQL> CREATE OR REPLACE FUNCTION hello_func
  2  RETURN VARCHAR2
  3  IS
  4  BEGIN
  5      RETURN 'Hello World';
  6  END;
  7  /

Function created.

SQL>
SQL> DECLARE
  2      msg VARCHAR2(20);
  3  BEGIN
  4      msg := hello_func();
  5      DBMS_OUTPUT.PUT_LINE(msg);
  6  END;
  7  /
Hello World

PL/SQL procedure successfully completed.
```

---

11. Function to return total number of employees

```
SQL> -- Q11: Function to count employees
SQL>
SQL> CREATE OR REPLACE FUNCTION totalEmployees
  2  RETURN NUMBER
  3  IS
  4      total NUMBER;
  5  BEGIN
  6      SELECT COUNT(*)
  7      INTO total
  8      FROM EMPLOYEE_24BYB1117;
  9
 10      RETURN total;
 11  END;
 12  /

Function created.
```

---

12. Function to display course description

```
SQL> CREATE OR REPLACE FUNCTION show_description
  2  (cno COURSE_24BYB1117.COURSE_NO%TYPE)
  3  RETURN VARCHAR2
  4  IS
  5      descp VARCHAR2(50);
  6  BEGIN
  7      SELECT description
  8      INTO descp
  9      FROM COURSE_24BYB1117
 10      WHERE course_no=cno;
 11
 12      RETURN descp;
 13
 14  EXCEPTION
 15      WHEN NO_DATA_FOUND THEN
 16          RETURN 'Course not available';
 17  END;
 18  /
```

13. Procedure to print Hello World

```
SQL> -- Q13: Procedure to print Hello World
SQL>
SQL> CREATE OR REPLACE PROCEDURE procHello
  2  IS
  3  BEGIN
  4      DBMS_OUTPUT.PUT_LINE('Hello World');
  5  END;
  6  /

Procedure created.
```

```
SQL>
SQL> EXEC procHello;
Hello World
```

14. Procedure with IN parameter

```
SQL> -- Q14: Procedure with IN parameter
SQL>
SQL> CREATE OR REPLACE PROCEDURE greet_user(name IN VARCHAR2)
  2  IS
  3  BEGIN
  4      DBMS_OUTPUT.PUT_LINE('Welcome ' || name);
  5  END;
  6  /

Procedure created.

SQL>
SQL> EXEC greet_user('Rakshit');
Welcome Rakshit
```

15. Procedure with OUT parameter

```
SQL> -- Q15: Procedure using OUT parameter
SQL>
SQL> CREATE OR REPLACE PROCEDURE greet_out(msg OUT VARCHAR2)
  2  IS
  3  BEGIN
  4      msg := 'Hello from procedure';
  5  END;
  6  /

Procedure created.

SQL>
SQL> DECLARE
  2      v_msg VARCHAR2(50);
  3  BEGIN
  4      greet_out(v_msg);
  5      DBMS_OUTPUT.PUT_LINE(v_msg);
  6  END;
  7  /
Hello from procedure

PL/SQL procedure successfully completed.
```

---

16. Procedure with IN OUT parameter

```
SQL> -- Q16: Procedure using IN OUT parameter
SQL>
SQL> CREATE OR REPLACE PROCEDURE greet_inout(p IN OUT VARCHAR2)
  2  IS
  3  BEGIN
  4      p := 'Hello ' || p;
  5  END;
  6  /

Procedure created.

SQL>
SQL> DECLARE
  2      name VARCHAR2(50) := 'Rakshit';
  3  BEGIN
  4      greet_inout(name);
  5      DBMS_OUTPUT.PUT_LINE(name);
  6  END;
  7  /
Hello Rakshit

PL/SQL procedure successfully completed.
```

17. Procedure to compute employee bonus

```
SQL> -- Q17: Procedure to compute employee bonus
SQL>
SQL> CREATE OR REPLACE PROCEDURE calc_bonus
  2  (empid IN NUMBER, bonus OUT NUMBER)
  3  IS
  4  BEGIN
  5      SELECT salary*0.10
  6      INTO bonus
  7      FROM EMPLOYEE_24BYB1117
  8      WHERE EMP_ID=empid;
  9  END;
 10  /

Procedure created.
```

18. Procedure to insert values into student table

```
SQL> -- Q18: Insert values into STUDENT table
SQL>
SQL> CREATE OR REPLACE PROCEDURE insert_student
  2  (id NUMBER, name VARCHAR2, dept VARCHAR2)
  3  IS
  4  BEGIN
  5      INSERT INTO STUDENT_24BYB1117 VALUES(id,name,dept);
  6      COMMIT;
  7  END;
  8  /
```

19. Procedure to check if employee exists

```
SQL> -- Q19: Check employee existence
SQL>
SQL> CREATE OR REPLACE PROCEDURE check_employee(eid NUMBER)
  2  IS
  3      cnt NUMBER;
  4  BEGIN
  5      SELECT COUNT(*)
  6      INTO cnt
  7      FROM EMPLOYEE_24BYB1117
  8      WHERE EMP_ID=eid;
  9
 10      IF cnt=0 THEN
 11          RAISE_APPLICATION_ERROR(-20001,'Employee does not exist');
 12      ELSE
 13          DBMS_OUTPUT.PUT_LINE('Employee exists');
 14      END IF;
 15  END;
 16  /

Procedure created.
```

---

20. Example of exception handling

```
SQL> DECLARE
  2      v_name EMPLOYEE_24BYB1117.EMP_NAME%TYPE;
  3  BEGIN
  4      SELECT EMP_NAME
  5      INTO v_name
  6      FROM EMPLOYEE_24BYB1117
  7      WHERE EMP_ID=&ID;
  8
  9      DBMS_OUTPUT.PUT_LINE(v_name);
 10
 11  EXCEPTION
 12      WHEN NO_DATA_FOUND THEN
 13          DBMS_OUTPUT.PUT_LINE('No such employee');
 14  END;
 15  /
Enter value for id: 1
old   7:     WHERE EMP_ID=&ID;
new   7:     WHERE EMP_ID=1;
Anand

PL/SQL procedure successfully completed.
```